Output

```
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit Program
Please Enter your choice: 1
----------- Input Data -----------
please enter number of rows for array: 2
please enter number of column of array: 2
please enter 1 column value: 12
please enter 2 column value: 34
please enter 1 column value: 56
please enter 2 column value: 78
----------- Main Menu -----------
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit Program
Please Enter your choice: 2
[12, 34]
[56, 78]
Compute total elements and mean using global variable.
total_elements : 4
Average : 45.0
```

```
----------- Main Menu -----------
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit Program
Please Enter your choice: 6
return maximum value, minimum value, total of the list values
----------- Data Summary -----------
values using *args:
Min: 12
Max: 78
Total: 180
total_elements : 4
Average : 45.0
```

```
----------- Main Menu -----------
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit Program
Please Enter your choice: 3
Please enter number to calulate its factorial: 5
Factorial of the 5 is:  120
----------- Main Menu -----------
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit Program
Please Enter your choice: 4
Filter 2D array values greater than given threshold value using lambda and filter function
Enter threshold value: 40
Filtered Data:  [56, 78]
```

```
----------- Main Menu -----------
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit Program
Please Enter your choice: 5
Sort 2D data using sorted()
Choose Sorting option:
1. Ascending
2. Descending
Enter your choice: 1
Sorted Data in Ascending order:  <function sort_data at 0x10396b3d0>
12, 34, 56, 78
----------- Main Menu -----------
1. Input Data
2. Display Data Summary
3. Calculate Factorial
4. Filter Data by Threshold
5. Sort Data
6. Display Dataset Statistics
7. Exit Program
Please Enter your choice: 5
Sort 2D data using sorted()
Choose Sorting option:
1. Ascending
2. Descending
Enter your choice: 2
sorted Data in Decending order:  <function sort_data at 0x10396b3d0>
78, 56, 34, 12
```